

SNR. OHANGA MAX



0728217318

KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE FOODS,
NUTRITION AND DIETETICS
HFN 200/280: HUMAN ANATOMY AND PHYSIOLOGY

1ST SEMESTER 2025/2026

TIME: 2 HOURS

INSTRUCTIONS: Answer all questions in both sections

SECTION A (30 MARKS)

1. Explain the following as applied in human physiology.
 - a) Extra cellular fluid. (4 marks)
 - b) Intra cellular fluid (3 marks)
 - c) Osmosis (3 marks)
2. Explain six functions of epithelial tissues in human body. (6 marks)
3. Describe the role of the adrenal glands in regulation of the human body metabolism. (4 marks)
4. Using diagrams where applicable, illustrate the structural and functional differences between various components of the human blood. (10 marks)

SECTION B (30 MARKS)

5. Using diagrams, illustrate how the skin achieves its various functions of maintenance of body homeostasis. (10 marks)
6. Describe the physiological functions of the following organs of the human body.
 - a) Ovaries (5 marks)
 - b) Testis (5 marks)
7. a) Draw a well labelled diagram of the human mammary gland. (5 marks)
b) Describe the hormonal control of lactogenesis in a breastfeeding mother. (5 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

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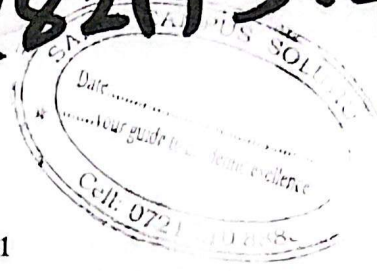
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HFN 280 0728217318



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE FOODS, NUTRITION AND DIETETICS

HFN 245: HUMAN ANATOMY

DATE: Thursday, 8th April 2021

HFN 280

TIME: 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS:

- Answer ALL Questions in both Sections.
- Start each question on a new page.

1. Describe the adaptations of the following in human body.
 - a) Synovial joints. (6 marks)
 - b) Stomach (6 marks)
2. Describe how the following body systems are involved in maintaining the body homeostatic balance on an extremely hot day.
 - a) Integumentary system (7 marks)
 - b) Urinary system (7 marks)

SECTION B

3. Draw clear well-labeled diagram of the following human organs.
 - a) Ovary (8 marks)
 - b) Heart (8 marks)
 - c) Eye (8 marks)
4. The lymphatic system plays a very important role in regulation of body fluids.
 - a) Using a simple well-labeled diagram of a lymphatic capillary, explain how they collect excess extracellular fluids. (6 marks)
 - b) Using Starlings Law, describe how the lymphatic system maintains the volume of the blood volume constant. (6 marks)
 - c) Discuss how the lymphatic system achieves its role of defense of the body against invading microorganisms. (8 marks)

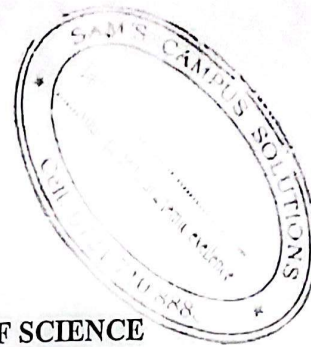
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SNR. OHANGA



HFN 245



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

(FOOD, NUTRITION AND DIETETICS)

HFN 245: HUMAN ANATOMY

1ST SEMESTER 2024/2025

TIME: 2 HOURS

INSTRUCTIONS: ANSWER ALL QUESTIONS. START EACH QUESTION ON A NEW PAGE.

SECTION A: (30 MARKS)

1. Explain features of THREE kinds of human striated muscles and their functions. (6 marks)
2. Describe how the following human organs are structurally adapted to their functions:
 - (a) Uterus (5 marks)
 - (b) Small intestines (5 marks)
3. Explain the functions of the various parts of the human lungs. (4 marks)
4. Describe the functions of the following organs in the human body:
 - (a) Thyroid glands (5 marks)
 - (b) Ear (5 marks)

SECTION B: (40 MARKS)

5.
 - (a) Draw a well labeled diagram of an adult human female breast. (10 marks)
 - (b) Explain how various parts of the above human organ function together to accomplish its role inn breastfeeding. (10 marks)
6.
 - (a) Draw a well labeled diagram of internal structure of the human heart. (10 marks)
 - (b) Describe how the above body organ accomplishes its role in blood circulation. (10 marks)

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HFN

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KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

(FOOD, NUTRITION AND DIETETICS)

HFN 200: HUMAN ANATOMY AND PHYSIOLOGYDATE: FRIDAY 11TH JUNE 2021

TIME: 8.00 A.M - 10.00 A.M

INSTRUCTIONS:

Answer all questions in both sections. Start each question on a new page.

1. Describe the following as applied in human anatomy and physiology.
 - a) Body fluids (4 marks)
 - b) Stratified squamous epithelia tissues. (4 marks)
 - c) Simple squamous epithelia tissues. (4 marks)
2. Draw and label the following parts of a human body.
 - a) Long bone (4 marks)
 - b) Skeletal muscle unit (4 marks)
 - c) Neuron (4 marks)
3. Describe FOUR main functions of the following body systems.
 - a) Endocrine system (8 marks)
 - b) Lymphatic system (8 marks)

SECTION B

4. The human kidney is an important organ in the regulation of body homeostatic balance. describe on how it achieves its function. (6mks)
5. Using well labeled diagrams describe
 - a) Mammary Glands (8 marks)
 - b) Skin (8 marks)
 - c) Lungs (8 marks)

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HFN 280



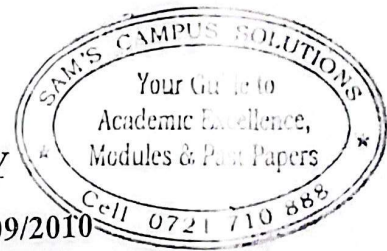
KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2009/2010

INSTITUTIONAL BASED EXAMINATION FOR THE DEGREE OF

BACHELOR OF EDUCATION (HOME ECONOMICS)

HFN 200: HUMAN ANATOMY AND PHYSIOLOGY



DATE: Friday 30th April, 2010

TIME: 8.00 a.m. – 10.00 a.m.

SECTION A – Answer *ALL* questions.

- Q1. List the characteristics of the erythrocytes (red blood cells). [5 marks]
- Q2. Describe the Cardiac Cycle. [5 marks]
- Q3. Draw a cross section of the Kidney and label the parts. [5 marks]
- Q4. Explain one cause of infertility in men. [5 marks]
- Q5. Explain the hormonal control of breast milk production. [5 marks]
- Q6. Explain the differences between hormonal and nervous controls. [5 marks]

SECTION B – Answer *TWO* questions only.

- Q7. (a) Draw a labeled diagram of the human digestive system. [10 marks]
(b) Explain the digestive processes that take place after eating a meal predominantly protein. [10 marks]
 - Q8. (a) Draw a labeled structure of the femur bone. [8 marks]
(b) Explain the causes of osteoporosis. [12 marks]
 - Q9. Write explanatory notes on each of the following:
(a) Eye defects. [5 marks]
(b) Action of progesterone and oestrogen hormones. [5 marks]
(c) Mitosis cell division and its significance. [5 marks]
(d) Gaseous exchange in the lungs. [5 marks]
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AFN 200/280



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATION 2011/2012 ACADEMIC YEAR

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCINENCE

HFN 200: HUMAN ANATOMY AND PHYSIOLOGY

DATE: Wednesday, 7th December, 2011

TIME: 11.00a.m. – 1.00p.m.

INSTRUCTIONS:

Answer ALL Questions

1. State **two** functions of the following organelles. [10 Marks]
 - (i) Mitochondria [2 Marks]
 - (ii) Lysozyme [2 Marks]
 - (iii) Golgi complex [2 Marks]
 - (iv) Ribosome [2 Marks]
 - (v) Nucleus [2 Marks]
2. State and describe causes of two main erythrocyte disorders. [10 Marks]
3. Describe **two** main kinds of salivary glands and in each case state **three** specific types and one function of these glands in food digestion. [10 Marks]
4. Differentiate the following terms as used in human anatomy and physiology [10 Marks]
 - (a) Simple diffusion and facilitated diffusion.
 - (b) Mitosis and meiosis.
 - (c) Simple epithelia and stratified epithelia.
 - (d) Universal donor and universal receptor.

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- Q 22/002 4 3 3
- (e) Lymphocytes and osteoblasts.
 - (f) Mastication and ingestion.
 - (g) Sprain and strain.
 - (h) Osteoporosis and arthritis.
 - (i) Endocrine and exocrine glands.
 - (j) Metabolic water and preformed water.



5. With the aid of a diagram describe at least five physiology of respiratory organs. [10 Marks]
6. Explain five similarities between nervous and endocrine system. [10 Marks]
7. Identify specific hormones describing how they complement the kidney in controlling the following process
- (1) Blood pressure
 - (2) Blood sugar level control
 - (3) Edema
 - (4) Dehydration
 - (5) Psychological stress control. [10 Marks]

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